From data to transfer decisions

Leicester City, summer 2016. Replace N'Golo Kanté, rebuild the spine, keep the budget.

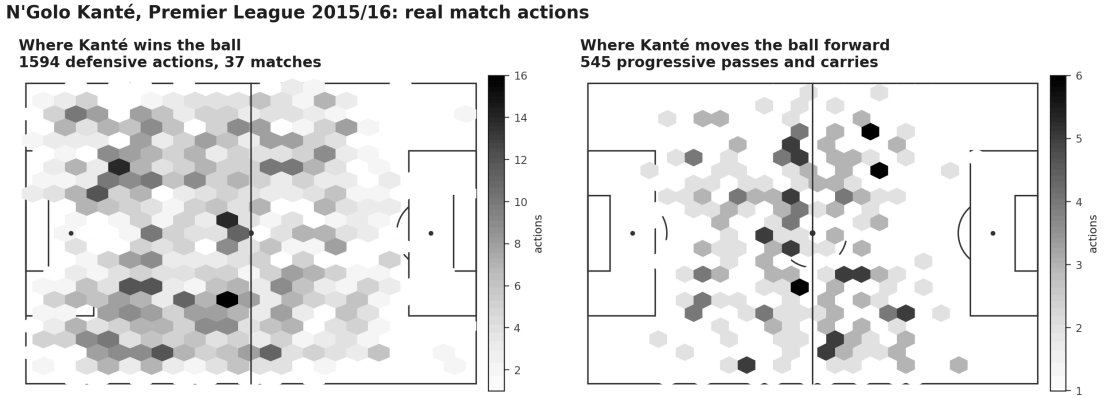
Dinmukhammed Mynzhassar, Laura Duisenova, Zhaidar Kairat

AAI-2504M

380
real matches parsed

30,939
player seasons priced

3
signings recommended



Replace Kanté without losing the identity

Sign Gueye, Cook and Ighalo for EUR 24m

CM

Idrissa Gueye

Aston Villa

EUR 9m

Fit 95 I Safety 91

CB

Steve Cook

AFC Bournemouth

EUR 3m

Value 100 I Safety 96

ST

Odion Ighalo

Watford

EUR 12m

Fit 79 I Value 82

EUR 50m budget EUR 26m left for wages and a January buffer Two year value backtest: +52%

The case for Gueye is tactical fit, not a bargain. The money is made on Cook. Ighalo is the best available box threat.

What the data can see

Two independent sources, one cutoff date: 1 July 2016.

380
Premier League matches

10,385
player-match rows

30,939
top five player seasons

1.1m
transfer records

01
Parse events
Passes, carries, shots, pressures, defensive actions

02
Build profiles
Per-90 rates, role clusters, similarity

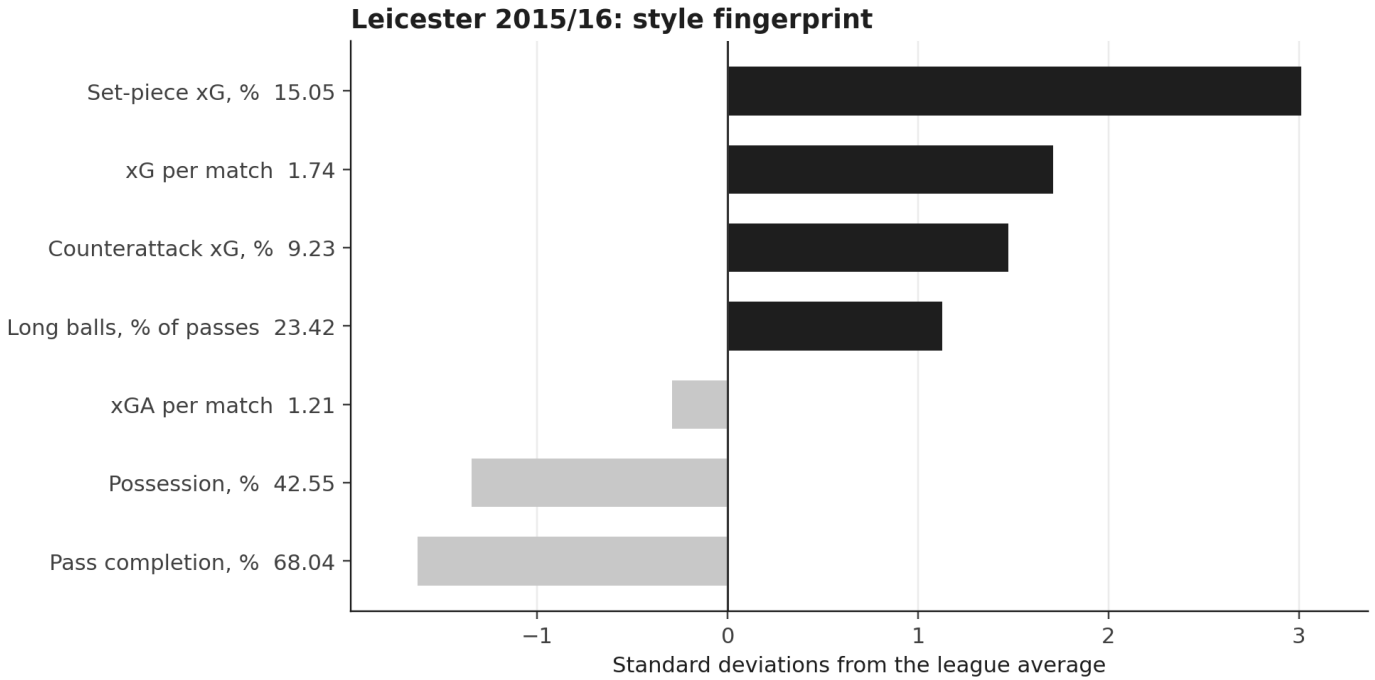
03
Price the market
Global and local value models

04
Score fits
Kanté similarity, need profiles, risk

05
Optimise
Binary plan under the fee budget

Raw data is not committed. The download script and processed tables make the result reproducible.

What Leicester were



League average is zero. One unit is one standard deviation across 20 teams.

42.6%
possession

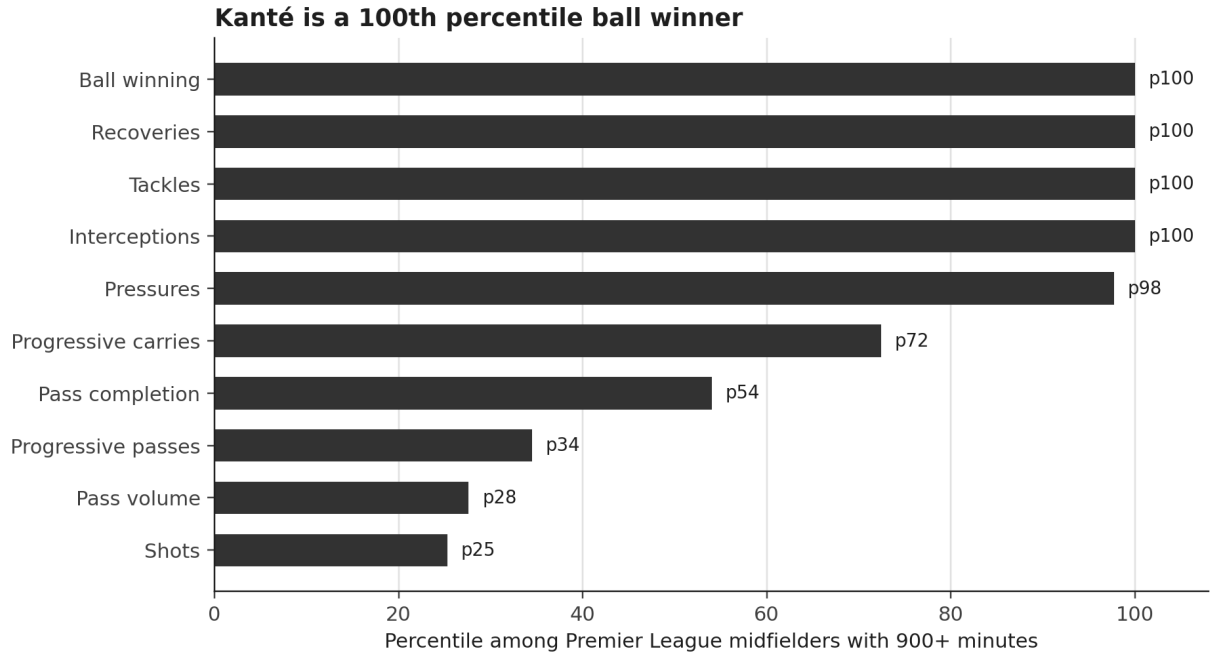
68.0%
pass completion

1.74
xG per match

15.1%
set-piece xG share

Low possession, direct passing, high xG, set pieces and counters. The style did not need a playmaker. It needed a destroyer who could carry the ball.

Kanté was not a playmaker



Kanté was a 100th percentile ball winner. He was not a high-volume passer.

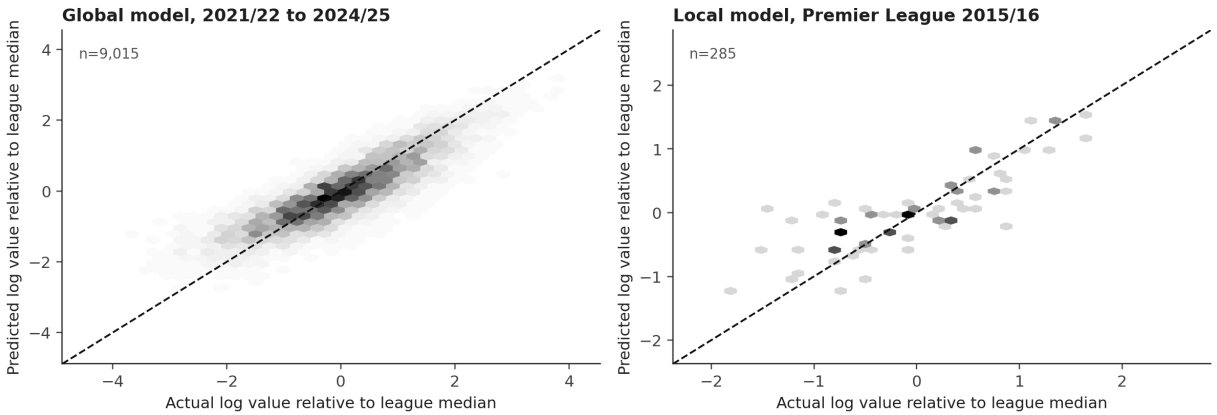
Cosine similarity to Kanté

0.93	Idrissa Gueye
0.75	Jan Kirchhoff
0.70	Victor Wanyama
0.68	Francis Coquelin
0.00	Aaron Ramsey, a label warning

Cosine similarity on ten per-90 actions. Formation labels are not roles.

Two market models

Market value models, out of sample



0.734
global R2

0.855
global Spearman

0.614
local R2

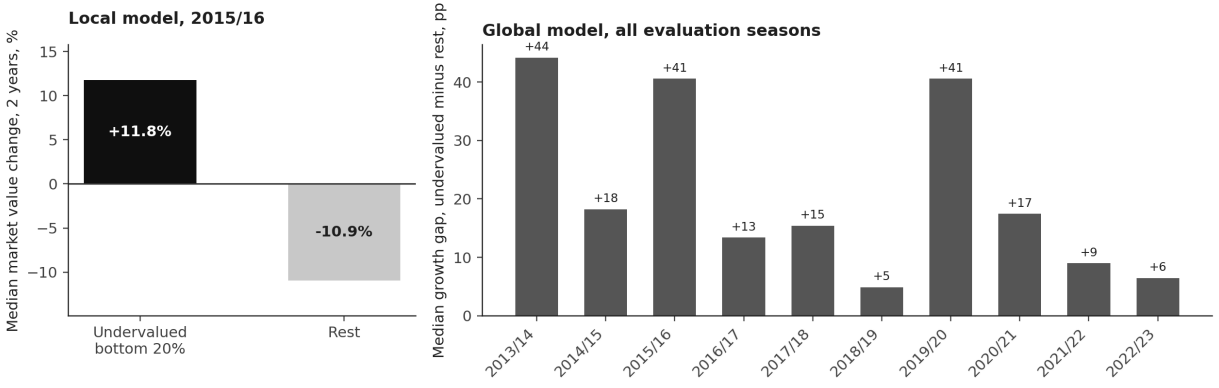
0.785
local Spearman

The market pays for goals, assists, passing volume and club reputation. It discounts defensive actions once position and possession are controlled.

Local baseline R2 0.495. The event model adds 0.119. Global test seasons: 2021/22 to 2024/25.

The undervalued signal is modest but real

The undervalued signal holds in relative terms



+11.8%
undervalued, 2 years

-10.9%
rest, 2 years

52.8%
hit rate, undervalued

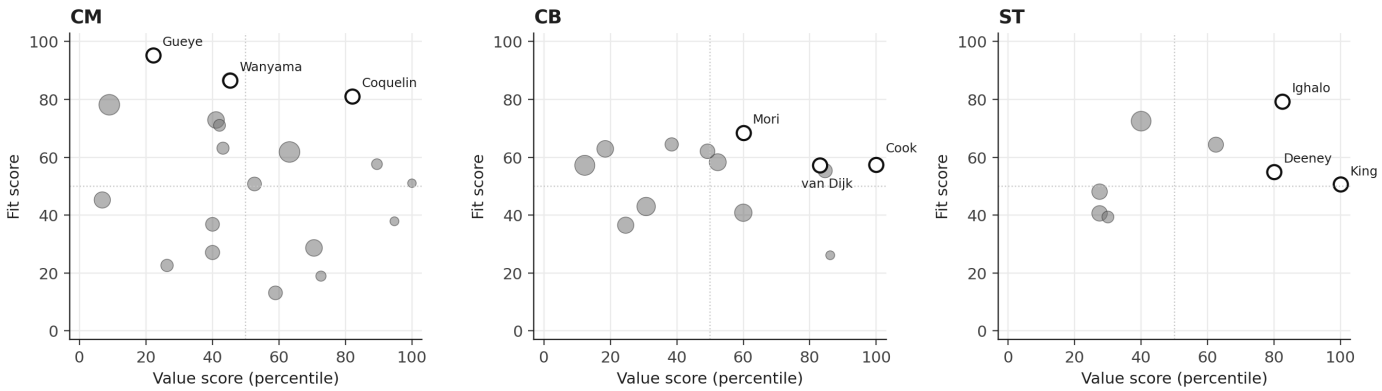
35.8%
hit rate, rest

The global model beat the rest in all 10 evaluation seasons on a relative basis. Absolute values fell in several seasons.

This is a signal, not a target. Individual outcomes vary.

The shortlist

Candidate map: tactical fit against market value, point size is fee



Need	Best balanced	Fee	Fit	Value	Safety	Score
Midfield	Francis Coquelin	EUR 12.0m	81	82	74	80
Defence	Steve Cook	EUR 3.0m	57	100	96	76
Attack	Odion Ighalo	EUR 12.0m	79	82	96	83

The shortlist is a trade-off surface, not a single ranking. Cook is the value extreme. Gueye is the fit extreme. Coquelin is the balance.

The plan

CM

Idrissa Gueye

Aston Villa

EUR 9m

Fit 95 I Value 22 I Safety 91

CB

Steve Cook

AFC Bournemouth

EUR 3m

Fit 57 I Value 100 I Safety 96

ST

Odion Ighalo

Watford

EUR 12m

Fit 79 I Value 82 I Safety 96

Total EUR 24m Left EUR 26m Age 25 to 27 All three needs filled

+80%
Gueye, 2016 to 2018

+85%
Cook, 2016 to 2018

-23%
Ighalo, 2016 to 2018

Backtest, not a promise. Two hits and one miss. The plan leaves room for wages, agent fees and a January move.

Risks and what would change

What could go wrong

- **Gueye's passing.** A safe short passer, not a creative progressor. Keep Drinkwater as the passer.
- **Cook's step up.** Bournemouth defended deep. Pair him with a progressive partner.
- **Ighalo's outlier.** One strong season at 27. Use him as rotation, not a long contract.
- **Adaptation.** The 4-4-2 pressing triggers need pre-season work.

What would change the answer

- Budget minus 20%: the recommended plan still fits at EUR 24m.
- Budget minus 40%: the base plan still fits at EUR 27m.
- If the board wants a higher ceiling, Cook can become van Dijk for EUR 33m total.
- If the board wants more passing, Gueye can become Coquelin for EUR 27m.

Limitations: one league of event data, market value as a fee proxy, retrospective outcomes, noisy residuals.

A decision system, not a black box

It does not pick a player. It makes the trade-offs visible.

Reproduce

python scripts/download_data.py
python scripts/build_all.py
python -m streamlit run app.py

Verify

9 unit tests
Brute-force optimum matches MILP
All report numbers generated from scripts

Decide

Fit, value and safety are explicit.
The weights are a choice.
The cash buffer is a decision.

AI was used for scaffolding and drafting. The author checked the mapping, the model specification and the recommendation.